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M-403 RAW WATER SYSTEM

1. SCOPE OF WORK

The Bidder shall supply the complete Raw Water Tank, Service Water System and water supply system in building in accordance with part 4, subsection 1.2 “Summary of Work”.

Two (2) Raw Water Tanks shall be provided complete with accessories, including but shall not be limited to the following components.

- Tank
- Nozzles
- Manholes
- Spiral stairway, ladder and handrail
- Valves, piping and flexible joint
- Pumps and Motors
- Foundation bolts and nuts
- Instrumentation and control equipment
- Electrical equipment
- Painting
- Cables
- Electrical equipment

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4. Subsection 2.5 “Codes and Standards”.

The codes and standards extracted from the Part 4, Subsection 2.5 “Codes and Standards” are specifically applicable to the design manufacture and testing of the equipment included in this Specification.

AWWA	American Water Works Association
AISC	American Institute of Steel Construction
AISI	American Iron and Steel Institute
ANSI	American National Standards Institute
ASTM	American Society for testing and Material
API	American Petroleum Institute
IEEE	Institute of Electrical and Electronics Engineers
NEMA	National Electrical Manufacturers Association
SSPC	Structural Steel painting Council
NFPA	National Fire Protection Association

3. DESIGN AND OPERATING CONDITIONS

For Combined Cycle power plant, the raw water produced by Desalination plant shall be fed into the raw water storage tanks. The water shall be used for service water, fire fighting water and water supply system.

The installation of two (2) x 1500 m³ raw water storage tank shall be completed before the date when the steam turbine in operation.

The tank shall be a vertical, cylindrical, cone roof tank and with capacity sufficiency for the requirement of the plant, including fire water requirements in accordance with NFPA.

Two (2) x 100% service water pumps for each tank shall be installed.

Service water pump capacity approximately 2 m³/h.

The services water pumps shall transfer the raw water to water supply system, waste water treatment plant circulation water pump area and other miscellaneous area.

The water supply system shall include the automatic chlorine injection and control system and after treated water from elevated water tank.

Each service water pumps shall be provided with a continuous flow circulation line connected to the equipment to provide the minimum flow through the pumps. Minimum recirculation flow shall be added to the system requirements to determine design flow.

A level control valve shall be installed at the inlet of tank to maintain the tank water level.

Capacity of each pumps shall include the maximum requirement, these pump shall be arranged for automatic start and stop.

4. DETAILED REQUIREMENTS

4.1 Tank

The material of tank shall be carbon steel plate of welded construction designed and fabricated in accordance with API 650 including, appendix E, seismic Zone 4, and shall be supplied with pipe connections and manholes.

The plate thickness of the tank shell and roof shall be not less than 6.5 mm. Tank bottom course shall be at least 8 mm. Plate shall be ASTM A283 Grade C.

The tank shall be provided with two manholes, one manhole shall be located on the tank top and the other on the tank shell near the base.

The spiral stairway to the tank roof shall be installed, and a vertical ladder shall be installed inside the tank.

A handrail shall be installed on the roof.

The grounding wiring shall be installed.

Anchor bolts shall be provided if necessary, according to the design conditions.

The air vent shall be of a sufficient size and shall be installed.

4.2 Pump

Two (2) x 100% Capacity Horizontal centrifugal type pumps shall be installed in accordance with Part 4, Technical Specification M-006.

Pumps shall be installed out door at the pump station near the Raw Water Tanks.

The motor shall be in conformity with Electrical Technical Specification E-210.

4.3 Piping and Valves

The valves shall be provided for all nozzles.

The valves shall be of stainless steel or carbon steel with lining and of the 10 Bar rating flanged type.

The overflow piping shall be led to a drain trench.

Flexible joint of the stainless steel shall be installed on inlet and outlet piping of tank.

Level control valve with stop valves shall be installed on the inlet piping of tank, and shall be in conformity with Part 4 Technical Specifications I-001 “Instrumentation and Controls”

The piping shall be in conformity with Part 4, Technical Specification M-001 Piping.

The Valves shall be in accordance with Part 4, Technical Specifications M-004 and M-005.

4.4 Chlorine Injection Facility

A chlorine injection facility shall be provided to prevent the water supply systems.

One (1) solution tank of more then fifteen (15) days capacity for chlorine injection facility shall be provided. The solution tank shall be of corrosive proof material.

Two (2) sets of chlorine injection pumps of 100% capacity for each plant shall be installed. The pump discharge shall be of the automatic control type. the liquid side of each pump shall be constructed of corrosion proof material. Chlorine injection pumps and accessories shall be per Part 4, Specification M-009.

Piping, valves, etc., shall be provided from the solution tank to the injection point.

4.5 Painting

Both inner and outer surfaces of all tank plates shall be sand blasted or shot blasted to remove mill scale, and shall be painted by one (1) layer of zinc-enriched primer before shipment.

After the installation of tank, the weld areas and rusted parts of shop painted plates shall be re- painted by primer paint after the removal of rust on scale by a grinder.

An epoxy resin based paint with minimum 300 micron in thickness shall be painted on the inner surface, and phenolated alkyd resin base paint with minimum 150 micron in thickness be painted on the external surface.

For further requirements refer also to the Part 4, Specifications M-001, S-420 and S-430.

5. PERFORMANCE TESTING AND GUARANTEES

The Bidder shall conduct performance testing in accordance with Part 4, Sub-section 2.6 Testing and Guarantees and shall guarantee the following system guarantees.

As minimum the Bidder shall perform all the test as required but not be limited to the followings :

- Leak testing
- Radiography
- Liquid penetrant examination
- Magnetic particle examination

The Bidder shall execute the following shop tests and shall submit the test reports to the Owner.

- (a) Material
- (b) Dimension check

The Bidder shall execute the following field tests and shall submit the test reports to the Owner.

- (a) Dimension check
- (b) Water filling test (hydrostatic test)
- (c) Penetrant and X-Ray test
- (d) Settlement check of tank base
- (e) Inspection for painting

6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality Assurance/Quality Control shall conform to the requirements of Part 4, Subsection 2.7.

The quality assurance/quality control levels for the tank shall correspond to the levels imposed on the equipment or system of which the respective tanks are part thereof.

The quality assurance/quality control procedures so specified in Part 4, Subsection 2.7 and shall include, but not limited to the following:

- Welding
- Nondestructive Testing
- Painting
- Packaging and shipping

The quality control documentation as specified in Part 4, Subsection 2.7 shall be as required by the applicable codes.

6.1 Radiography

All radiography shall be in accordance with the procedure as and comply to the acceptance standards referenced in API 650.

Spot radiography shall be modified as follows

One radiograph shall be taken in the first 3 m of weld in each vessel and one spot shall be examined for each additional 15 m of full-penetration butt welds, or fraction thereof.

At least 2 intersection of vertical and horizontal joints on each tank, when they exist, shall be included in the radiographs.

All films shall be subject to the engineer's review and comment.

6.2 Leak Testing

Each completed tank shall be leak-tested following all fabrication, but prior to painting.

The tank shall be filled with water and inspected frequently during filling.

If water is not available for testing, all joints shall be tested with a vacuum box and soap solutions. The differential pressure across the joints shall be at least 0.35 Bar

The bottom joints of flat-bottomed tanks shall be tested with a vacuum box and soap solution prior to hydrotesting.

Any leaks detected during hydro testing or vacuum testing shall be repaired by procedures approved by the owner.

The Bidder shall provide (and remove after use) necessary test pump and temporary piping, valves, blind flanges, etc. for filling and draining the tank.

Result of all tests shall be documented and submitted to the engineer with a sketch showing areas examined.

Test procedures shall be submitted for review by the Owner. No testing work shall be done until these procedures are acceptable to the owner.

7. INSTRUMENTATIONS AND CONTROLS

The Instruments and devices shall be provided by the Bidder but not limited to the following

The level transmitter and remote indicators provided in the Central Control Room and Demineralized water treatment system control panel.

Dial type level indicator with high and low level alarm switch for annunciator on Main Control Room.

Instrument and control devices requirement for service water pump interlock, fire pump interlock and water treatment equipment interlock.

Refer to Part 4, Technical Specification I-001 Instrumentation and Control for design requirements for the instrument and Control System and devices and I-002 Local Control Panel for design requirements for Local Control Panel.

8. ELECTRICAL SPECIFICATIONS.

8.1 General

The Bidder shall furnish and install all electrical equipment and accessories required for the Raw Water System.

Those equipment shall include but not be limited to the followings :

- (a) Motors
- (b) Power panels
- (c) Conduits, cables and wiring.
- (d) All electrical control, monitoring and instrumentation devices, and other appurtenances required for this specification, M-403 shall be provided by the Bidder.

For general electrical requirements, see Part 4, Technical Specification E-001.

8.2 Motors

For detailed electrical requirements, see Part 4, Technical Specification E-210 for Low Voltage Motors.

8.3 Power Panels

For detailed electrical requirements, see Part 4, Technical Specification E-001 and E-280.

The panel shall have a degree of protection IPW 55 of IEC 60529.

8.4 Conduits, Cables and Wiring

All motors and electrical, control, monitoring and instrumentation devices furnished and installed by the Bidder shall be wired by him to the power and local control panel respectively.

The wiring between those panels shall also be included.

The required power, control, instrumentation and thermocouple extension cables shall be provided by the Bidder.

For detailed electrical requirements, see Part 4, Technical Specifications E-230, E-231 and E-232.

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